Strict ratio log-concavity of Sun's R-sequence

Independent working note — MathDB 331960

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Task 1: solve the published conjecture. Define

$$R_n = \sum_{k=0}^{n} \binom{n}{k}\binom{n+k}{k}\frac{1}{2k-1}, \qquad r_n = \frac{R_{n+1}}{R_n} \quad (n \geq 1).$$

Conjecture 4.7 of Brian Yi Sun's published paper [1] asserts that $\{r_n\}_{n\geq 4}$ is log-concave. We prove the strict assertion, with an exact finite algebraic certificate for the induction.

Theorem 1. *For every integer $n \geq 5$,*

$$r_n^2 > r_{n-1}r_{n+1}, \quad \textit{equivalently} \quad R_{n+1}^3 R_{n-1} - R_n^3 R_{n+2} > 0.$$

Consequently the published Conjecture 4.7 is true, and its starting index 4 cannot be decreased.

Indexing. The MathDB transcription defines its ratio as R_n/R_{n-1} while retaining the starting index 4. In that notation the correct starting index is 5. This note proves the original published conjecture; it does not count the transcription error as a refutation.

A recurrence derived from the definition. For $n \geq 1$, the terms $k = 0, 1$ give $R_n \geq n(n+1) - 1 > 0$. Formal power series give

$$\sum_{n\geq 0} R_n z^n = \frac{1}{1-z}\sum_{k\geq 0}\frac{\binom{2k}{k}}{2k-1}\left(\frac{z}{(1-z)^2}\right)^k = -\frac{\sqrt{1-6z+z^2}}{(1-z)^2} =: G(z).$$

Here the middle identity follows from the binomial expansion $\sum_{k\geq 0}\binom{2k}{k}u^k/(2k-1) = -\sqrt{1-4u}$. Differentiating gives

$$(1-z)(1-6z+z^2)G'(z) = (z^2-8z-1)G(z).$$

Comparison of coefficients yields, for $n \geq 0$,

$$(n+3)R_{n+3} - (7n+13)R_{n+2} + (7n+15)R_{n+1} - (n+1)R_n = 0. \tag{1}$$

Explicit bounds. Put $w = \sqrt{2}$, $\rho = 3 + 2w$, and

$$P_{\pm}(x) = 64x^4 - 96x^3 + (168 - 78w)x^2 + (273w - 537)x \pm 640,$$
$$L(x) = \frac{P_-(x)}{64x^4}, \qquad U(x) = \frac{P_+(x)}{64x^4}.$$

Equivalently,

$$L(x), U(x) = 1 - \frac{3}{2x} + \frac{21/8 - 39w/32}{x^2} + \frac{-537/64 + 273w/64}{x^3} \mp \frac{10}{x^4}.$$

For $q_n = r_n/\rho$, recurrence (1) becomes

$$q_{n+2} = T_n(q_n, q_{n+1}), \qquad T_x(a,b) = \frac{7x+13}{(x+3)\rho} - \frac{7x+15}{(x+3)\rho^2 b} + \frac{x+1}{(x+3)\rho^3 ab}. \tag{2}$$

Lemma 1. *For every integer $n \geq 20$, $L(n) < q_n < U(n)$. Moreover, for every real $x \geq 20$,*

$$L(x) > 0, \qquad (7x+15)\rho L(x) > x+1, \tag{3}$$
$$T_x(U(x), L(x+1)) > L(x+2), \tag{4}$$
$$T_x(L(x), U(x+1)) < U(x+2), \tag{5}$$
$$L(x)^2 > U(x-1)U(x+1). \tag{6}$$

Proof. We first verify the four real-variable assertions by exact polynomial arithmetic. The following five expressions are polynomials in $\mathbb{Z}[w][x]$:

$$\begin{aligned}
H_P(x) &= P_-(x), \\
H_M(x) &= (7x+15)\rho P_-(x) - 64(x+1)x^4, \\
H_L(x) &= 64(x+3)\rho^3(x+2)^4 P_+(x)P_-(x+1) \\
&\qquad \cdot \big[T_x(U(x), L(x+1)) - L(x+2)\big], \\
H_U(x) &= 64(x+3)\rho^3(x+2)^4 P_-(x)P_+(x+1) \\
&\qquad \cdot \big[U(x+2) - T_x(L(x), U(x+1))\big], \\
H_C(x) &= 4096x^8(x-1)^4(x+1)^4\big[L(x)^2 - U(x-1)U(x+1)\big].
\end{aligned}$$

Cancellation of the displayed denominators proves polynomiality. Write

$$H_J(20+y) = \sum_{k=0}^{d_J} (a_{J,k} + b_{J,k}w)y^k.$$

All $a_{J,k}, b_{J,k}$ are integers. Expansion using $w^2 = 2$ gives the following coefficient certificate:

J	d_J	$\min_{0\leq k\leq d_J} \min\{a_{J,k} + \frac{7}{5}b_{J,k}, a_{J,k} + \frac{3}{2}b_{J,k}\}$
P	4	64
M	5	12672/5
L	9	1574064128/5
U	9	2525868032/5
C	13	12288

Since $7/5 < w < 3/2$, every coefficient in each shifted polynomial is strictly positive. Hence all five $H_J(x)$ are positive for $x \geq 20$. The first two give (3). Since $P_+ = P_- + 1280$, the multipliers defining H_L, H_U, H_C are positive on this range. The other three give (4)–(6).

For clarity, the coefficient table is a finite exact certificate, not an asymptotic estimate. Its complete integer coefficient pairs are recorded in `exact-certificate.json`. Two independent implementations check them: `verify_proof.py` uses quadratic-field polynomial arithmetic, and

`verify_independent.py` expands the products using only integer and rational pairs and polynomial convolution. The latter needs only Python's standard library.

It remains to propagate the bounds along the actual sequence. Direct evaluation of the defining binomial sums gives

$$L(j) < q_j < U(j) \qquad (j = 20, 21).$$

The same certificate records all four positive gaps. For $a \geq L(n) > 0$ and $b > 0$, differentiation in (2) shows

$$\frac{\partial T_n}{\partial a} < 0, \qquad \frac{\partial T_n}{\partial b} = \frac{(7n+15)\rho a - (n+1)}{(n+3)\rho^3 a b^2} > 0,$$

where the second sign follows from (3). If the bounds hold at $n, n+1$, these signs imply

$$T_n(U(n), L(n+1)) \leq q_{n+2} \leq T_n(L(n), U(n+1)).$$

Equations (4) and (5) therefore give the strict bounds at $n+2$. Induction starting with $20, 21$ completes the proof. □

Proof of Theorem 1. For $n \geq 21$, all three indices $n-1, n, n+1$ lie in the range of Lemma 1. Thus

$$q_n^2 > L(n)^2 > U(n-1)U(n+1) > q_{n-1}q_{n+1}.$$

Multiplication by ρ^2 proves the desired inequality. For $5 \leq n \leq 20$, direct integer evaluation gives

$$\min_{5 \leq n \leq 20} \left(R_{n+1}^3 R_{n-1} - R_n^3 R_{n+2}\right) = 4658935984 > 0.$$

Every one of these sixteen exact integers is recorded in the certificate. They are computed independently from the binomial sum and from (1), starting with $R_0 = -1, R_1 = 1, R_2 = 7$. This covers all $n \geq 5$. Finally $R_3 = 25, R_4 = 87, R_5 = 329, R_6 = 1359$, and

$$R_5^3 R_3 - R_4^3 R_6 = -4623352 < 0.$$

Hence the ratio sequence cannot be extended back from index 4 to index 3 while preserving log-concavity. □

Status and verification. The published full text retains Conjecture 4.7. Current MathDB discussion and solution fields, focused literature searches, and public claim catalogues were checked; no matching solution claim was located. The separate literature audit records the sources, timestamps, scope correction, and search limits. The proof above covers the complete published statement; the computations certify only its explicitly stated finite algebraic obligations.

References

[1] Brian Yi Sun, *On a ratio monotonicity conjecture of a new kind of numbers*, Journal of Inequalities and Applications **2018**, article 24. doi:10.1186/s13660-018-1614-1. Conjecture 4.7. Earlier version: arXiv:1512.01008.

Task 2: determine the onset for the parameter family. For $t > 0$, define

$$R_n(t) = \sum_{k=0}^{n} \binom{n}{k} \binom{n+k}{k} \frac{t^k}{2k-1}.$$

Determine the least $N(t)$ for which the terms and consecutive ratios are positive and the sequence

$$\{R_{n+1}(t)/R_n(t)\}_{n \geq N(t)}$$

is log-concave. Locate the parameter values at which $N(t)$ changes.

Task 3: find sharp constants in the ratio bounds. With

$$B(n) = 1 - \frac{3}{2n} + \frac{21/8 - 39\sqrt{2}/32}{n^2} + \frac{-537/64 + 273\sqrt{2}/64}{n^3},$$

determine the infimum and supremum of $n^4(q_n - B(n))$ over integers $n \geq 20$. Prove where the extrema occur, including any limiting extremum.

Task 4: quantify higher orders of log-monotonicity. For each integer $k \geq 2$, determine the least threshold N_k such that

$$(-1)^k \Delta^k \log R_n > 0 \qquad (n \geq N_k), \qquad \Delta f(n) = f(n+1) - f(n).$$

Obtain effective bounds for N_k as k grows.

Task 5: find a structural proof of the ratio inequality. Develop an integral representation, a continued fraction, or a positivity-preserving operation that explains the strict ratio log-concavity directly. Determine which other sequences with generating functions $-\sqrt{1 - az + z^2}/(1-z)^b$ admit the same argument.